Mapping Heat

Cross-city UHI prediction, blind target.

HEADLINE RESULT

Freetown blind target. F1 macro = 0.689. Train/Test gap = 0.003.

Team 4 · MBAN Elites

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Eight-minute executive presentation · Three cities · One pipeline

Three cities. Three climates. One generalization test.

Heat kills where cities are built wrong. GDP does not protect residents.

~739

Chicago, 1995 · 5-day heatwave

Excess deaths in a single U.S. city. The canonical study of heat as a social disaster.

[1] *Semenza et al. (1996)*

61,672

Europe, 2022 · heat-related deaths

A full medical system did not prevent a warming Europe from dying in summer.

[2] *Ballester et al. (2023)*

3 cities

Rio, Santiago, Freetown

Three climate zones, two labeled, one blind. Every target is extrapolation, not interpolation.

[3] *Sentinel-2 + OSM + competition labels*

The technical question: can a model trained on two cities predict a third it has never seen?

[1] *Semenza, J. C., et al. (1996). Heat-related deaths during the July 1995 heat wave in Chicago. New England Journal of Medicine, 335(2), 84 to 90.*

[2] *Ballester, J., et al. (2023). Heat-related mortality in Europe during the summer of 2022. Nature Medicine, 29(7), 1857 to 1866.*

Satellite bands, building polygons, ground labels

Three public inputs. One of the three target cities has no labels.

SOURCE 1

Sentinel-2 L2A imagery

Fifteen spectral bands at 10 m resolution.

Fifteen cloud-filtered scenes per city, compiled into a median mosaic.

Accessed via Microsoft Planetary Computer STAC.

SOURCE 2

OpenStreetMap footprints

Per-city building polygons (Chile 191 MB, Brazil 151 MB, Sierra Leone 36 MB).

Feeds seven geometry features: count, density, coverage ratio, mean size, shape complexity.

Sierra Leone OSM is five times sparser than Chile. A known bias.

SOURCE 3

Ground-truth labels

Per-pixel UHI class (Low, Medium, High) from the competition sponsor.

Available for Santiago (14k points) and Rio (21k points).

Freetown has coordinates but no labels. Blind target.

Five notebooks. One pipeline. One target at a time.

The same code runs for each target city. Only the training configuration changes.

NOTEBOOK 01
Scene search
STAC query: fifteen cloud-filtered Sentinel-2 scenes per city.

NOTEBOOK 02
Median mosaic
Fifteen-band per-pixel median. One TIFF per city.

NOTEBOOK 03
Freetown
Blind target. Trained on Santiago + Rio only.

NOTEBOOK 04
Santiago
Mixed training. All three cities in the training set.

NOTEBOOK 05
Rio de Janeiro
Mixed training. All three cities in the training set.

INSIDE EACH TARGET NOTEBOOK

Feature engineering. Thirty-two per-pixel features across five tiers.

Per-city z-score normalization and spatial-block cross-validation on 60 KMeans clusters.

Soft-voting ensemble: Random Forest, HistGradientBoosting, Logistic Regression at weights 2 : 2 : 1.

Phase 4A physics-anchored pseudo-labeling for target adaptation without ground truth.

Thirty-two features across five tiers

The eleven geo-aware context features are the breakthrough layer.

TIER 1	*NDVI · NDBI · NDWI*	
Classical spectral		3
TIER 2	*SAVI · EVI · BSI · UI · NDMI · NDVIre · MNDWI · B09*	
Advanced spectral		8
TIER 3	*urban_heat · veg_cooling · water_cooling · balance terms*	
Physical composites		6
TIER 4	*count · density · coverage · size · shape complexity*	
Building geometry (OSM)		7
TIER 5	*distance to coast · forest · center · local density · land-use class*	
Geo-aware context		11

THE BREAKTHROUGH

Geo-aware context

Eleven features encode where a pixel sits in its city: distance to coast, forest, and city center; local urban and vegetation fractions; one-hot land-use class.

Impact on Freetown honest F1:

0.58 → 0.689

before → after geo-aware

Geographic reasoning matters. Pure spectral features are not enough for cross-climate transfer.

Per-city normalization plus spatial-block cross-validation

The two design choices that make one model work across three climates.

01 · PER-CITY NORMALIZATION

Z-score every feature within each city.

Raw NDVI in Santiago is very different from raw NDVI in Freetown.

After z-scoring, the model learns 'hotter than this city's own mean' rather than absolute thresholds that break across climates.

Result:

One model. Three cities. No per-climate retraining.

02 · SPATIAL-BLOCK CROSS-VALIDATION

Sixty KMeans clusters as CV blocks.

A random pixel-level split leaks spatially adjacent pixels between train and test. The CV score looks fantastic. It lies.

We cluster pixels into sixty geographically contiguous KMeans blocks. StratifiedGroupKFold then holds out entire blocks at a time.

Result:

No spatial autocorrelation leak. CV scores match held-out reality.

These two choices together lift Freetown honest F1 by roughly ten points over the naive baseline.

Physics-anchored stratified pseudo-labeling

How Freetown gets training signal without a single ground-truth label.

01 RANK

Score every target-city pixel by a physical heat composite.

Rank the full target city by heat.

02 FORCE-BALANCE

Top 15% become pseudo-High. Bottom 15% become pseudo-Low. Middle 15% around the median become pseudo-Medium.

Physics picks the labels, not model confidence.

03 FILTER

Keep only pseudo-labels where the current model predicts that class with probability ≥ 0.20.

Sanity filter. Kills physically impossible labels.

04 RETRAIN

Add pseudo-labels at weight 0.30. Run for two rounds.

Class balance is enforced every round.

PSEUDO-LABELS GENERATED

Freetown 7,978 · Santiago 7,229 · Rio 13,786 (across two rounds, per city)

Physics picks the class balance by construction. Confidence-based pseudo-labels collapse to the majority class.

One honest scorecard, three cities

Lead with Freetown: blind target, honest F1 0.689, train/test gap 0.003.

Target	Scenario	Gap ★	Honest F1 ★	F1 vs labels	Predicted
Freetown	**Blind cross-city**	**0.003**	**0.689**	—	—
Santiago	Mixed training	**0.181**	**0.549**	0.576	31/34/35%
Rio de Janeiro	Mixed training	**0.177**	**0.625**	0.626	33/29/38%

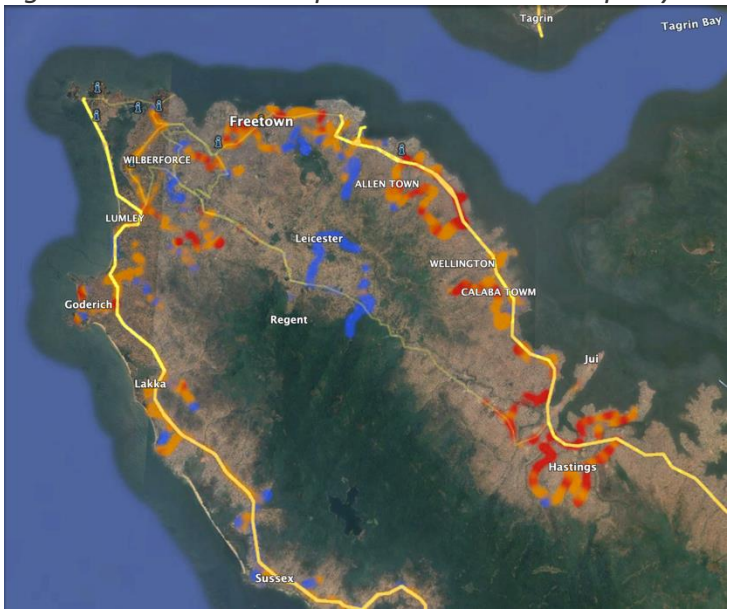
★ The two columns that matter. Small gap means honest generalization. Honest F1 is the target-city-only held-out blocks.

Santiago and Rio gaps (~0.18) are an expected artifact of mixed training, not overfitting. The Honest Generalization Eval cell strips it out.

Freetown is the cleanest test: no target labels seen during training, and no overfit signal on the spatial holdout.

Freetown: model predictions vs satellite physics

Agreement with an independent satellite heat proxy: 65.9%.



SATELLITE HEAT

NDBI minus NDVI. Red is hotter.

PREDICTION

Low / Medium / High per pixel.

OVERLAY

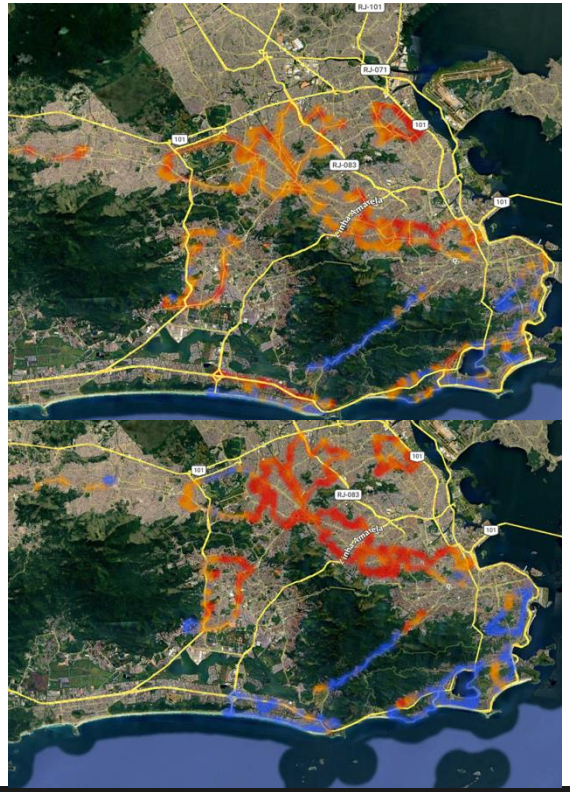
Prediction on top of satellite heat.

DISAGREEMENT

Red pixels: model and physics disagree.

Where we disagree concentrates on transitional Medium pixels, not on the urban / rural boundary.

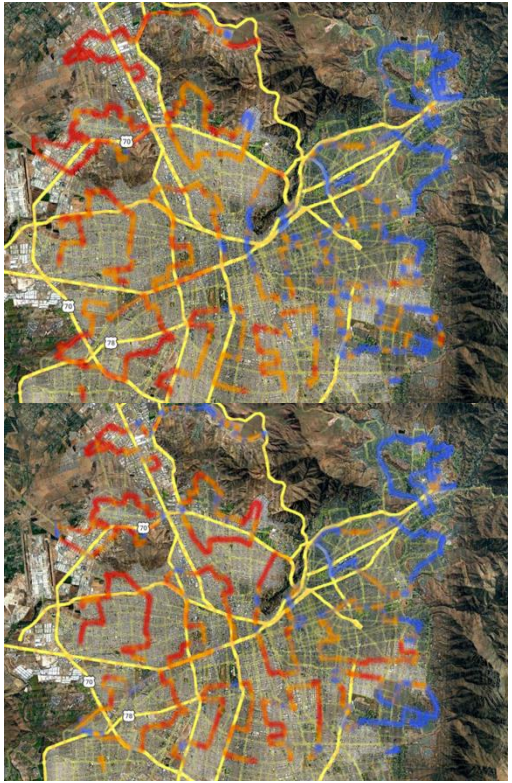
Santiago & Rio: model application & testing



Actual

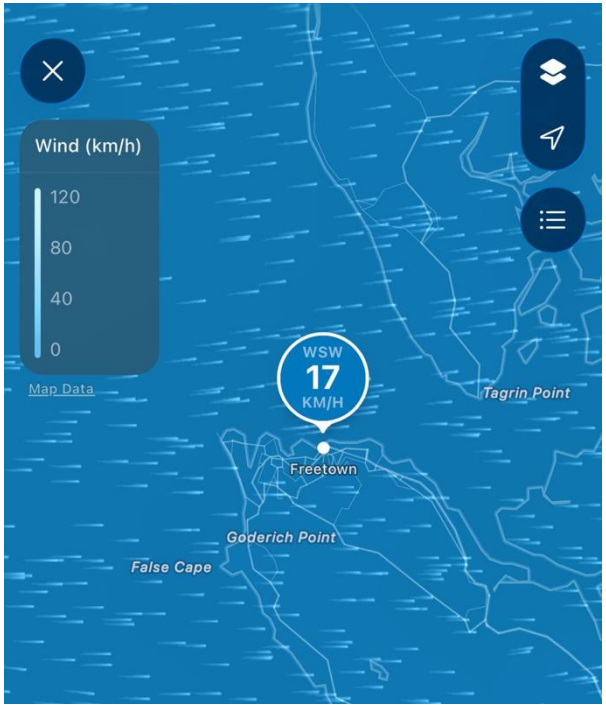
Vs

Prediction model



Where the model is trained on its own predicted dataset and preserves its intent to generalize raw data aiming min. gap, max F1 & accuracy.

Heat follows airflow, foehn & wind-corridor hypotheses



Temperature class may be decided by whether wind reaches a block — this is still a hypothesis.

HYPOTHESIS α

Foehn-override

Observation — Vegetated zones downwind of the eastern hills still register High UHI.

Interpretation — Mountain-lee descending flow may suppress evapotranspirative cooling.

Test — Local weather stations × wind-vector overlay.

HYPOTHESIS β

Wind-corridor × density

Observation — At equal built density, sea-breeze-facing blocks run cooler than sheltered blocks.

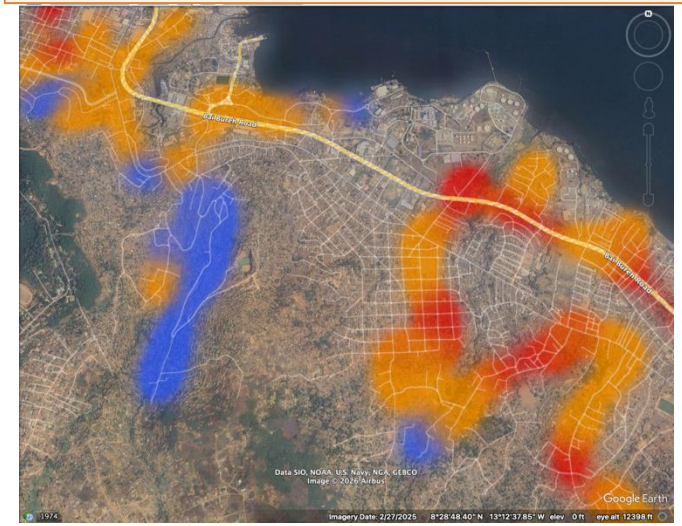
Interpretation — Street orientation and LiDAR heights may gate sea-breeze penetration.

Test — Building-height model × dominant street azimuth.

Heat follows form — impervious composition & intersection density

Urban form may outrank green cover — this is still a hypothesis.

LAYER 2 · Unvalidated observations — hypotheses for field testing



HYPOTHESIS γ

Impervious-surface composition

High-asphalt ratio associates with High-class pixels within 'built-up.'

HYPOTHESIS δ

Intersection micro-hotspots

Local High spots cluster at major-road intersections — signal-idling heat.

HYPOTHESIS ε

Urban form > green cover

Low-NDVI blocks with favorable canyon geometry stay in Medium.

If confirmed, inverts the 'plant more trees citywide' recommendation.

Mechanism hypotheses derived from Google Maps overlay inspection, not model output — ship as pilots, not policies.

What we ship, what we cannot, what comes next

Honest framing for an honest pipeline.

WHAT WE SHIP

1. Freetown predictions
Blind cross-city map with honest F1 0.689 and no overfit signal.

2. Open, reproducible pipeline
One conda environment, five notebooks, random seed 42, cached artifacts.

3. Executive report
Three-page summary for non-technical stakeholders and AI tools.

HONEST LIMITS

1. Medium is an in-between class
Precision drops on the narrow Medium slice. Expected, not a bug.

2. One dry-season snapshot
Season, wind, and time of day all change the heat pattern.

3. OSM coverage bias
Sierra Leone's building footprints are five times sparser than Chile's.

NEXT STEPS

1. Direct land-surface temperature
MODIS or Sentinel-3 SLSTR for a real Kelvin input. Expected plus five to ten points of F1.

2. More training cities
Five to eight climatically diverse cities turn extrapolation into interpolation.

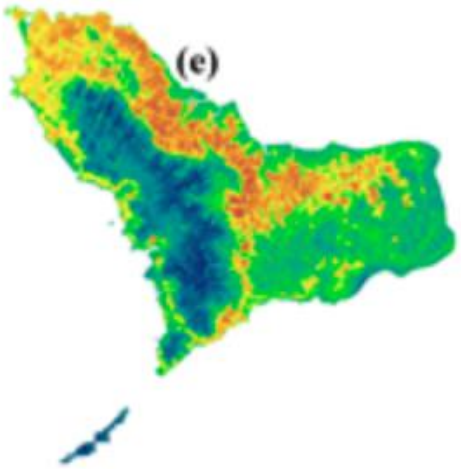
3. Google Open Buildings v3
Better informal-settlement coverage in African cities.

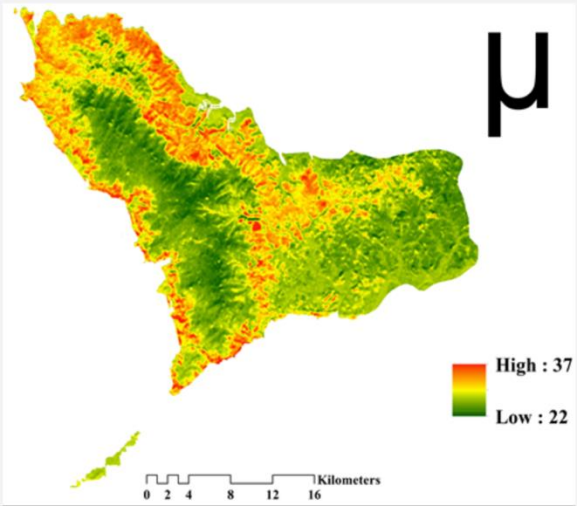
One pipeline. Three cities. An honest number and a clear path to a better one.

Curious? Scan Here!!!

FLOWCODE

PRIVACY.FLOWCODE.COM

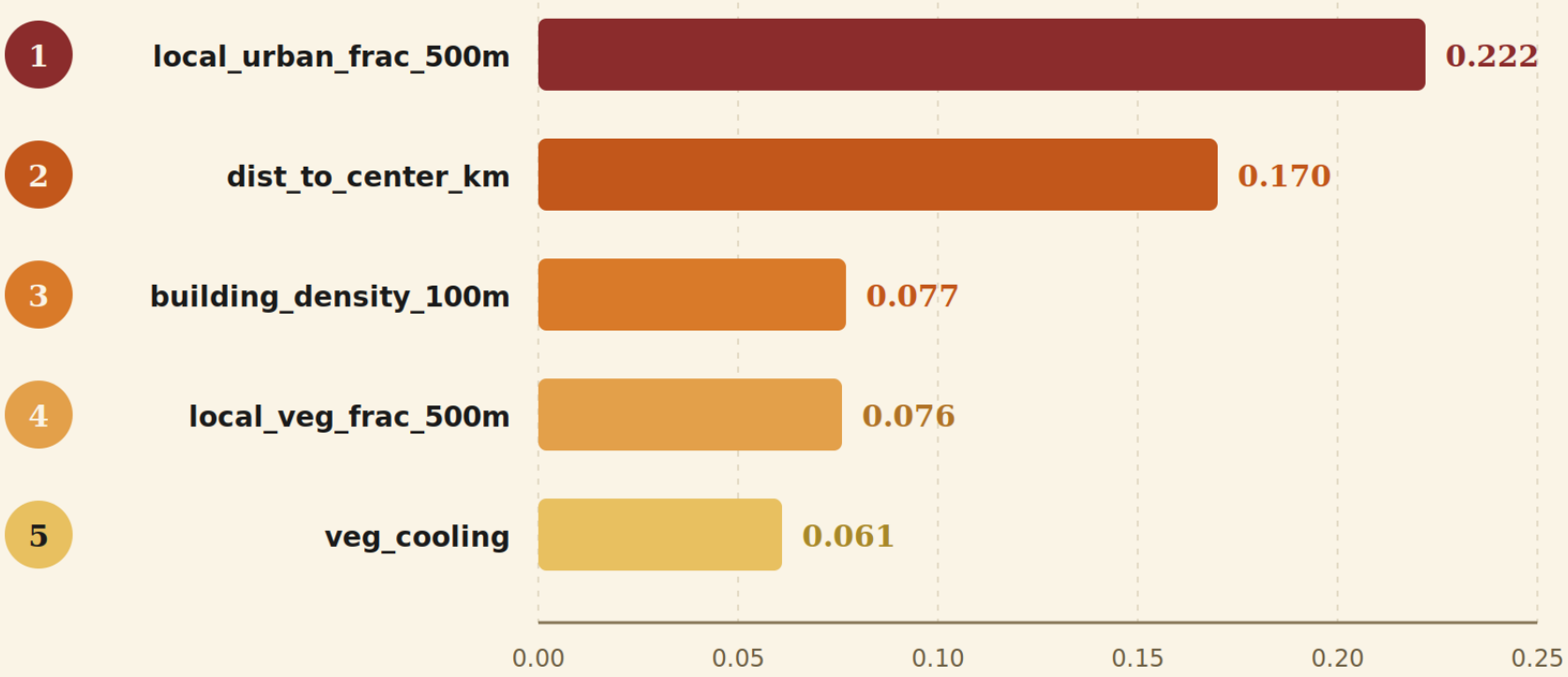




2018 **Our model** **2030**

Top 5 Features — Random Forest

SANTIAGO · RIO · FREETOWN



Urban form and distance-to-center dominate — the built environment is the signal.